

NATIONAL TELE-OPHTHALMOLOGY AI SCREENING PROGRAM

Explainable AI Clinical Decision Support System | Rural PHC Triage (SIH PS 26038)

Patient ID: IND-PHC-2026-9418 **Age/Sex:** 58 / Male **Screening Date:** 15-Sep-2026 09:20
PHC Health Centre: PHC Mulbagal, Kolar District #04 **Frontline Operator:** ASHA Health Worker #108
Eye Evaluated: Right Eye (OD) (Nasal: Left) **QA Usability Score:** 57/100 [WARNING]

TRIAGE ACTION: SECONDARY OPHTHALMOLOGY REFERRAL (WITHIN 30 DAYS)

Clinically significant Diabetic Retinopathy detected. Comprehensive dilated fundus exam advised.

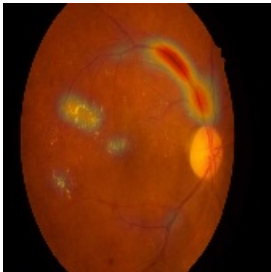
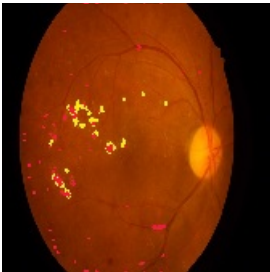
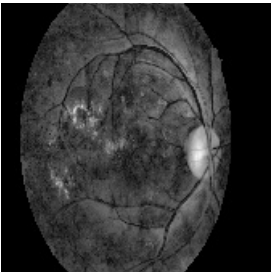
1. Raw Fundus (OD)

2. Enhanced Green

3. Retinal Vessels

4. Lesion Detections

5. Grad-CAM XAI

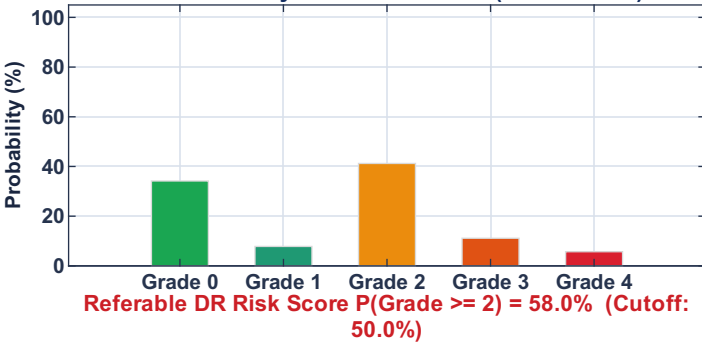


Legend: [Yellow]: Hard Exudates | [Ruby Red]: Microaneurysms & Hemorrhages | [Heatmap]: Deep Feature Attention (Grad-CAM)

QUANTITATIVE BIOMARKERS & ETDRS 4-2-1 CLINICAL AUDIT

ETDRS Staging: SEVERE NPDR (Criteria met: 4-2-1 Rule positive)
ETDRS 4-2-1 Rule: 4Q Hemo: [2/4] | VB Quads: [0] | IRMA: [1]
Distance to Fovea (CSME): 1711 μ m (1.14 DD) — Moderate
Hard Exudates (Bright): 26 clusters (999 px area)
Microaneurysms / Blot Hemo: 55 MAs | 4 Hemorrhages
Vascular Density: 9.52% FOV **Optic Disc:** [172, 226] (Left)
Systemic EHR Profile: T2D: 8 yrs | Last HbA1c: 8.2% | BP: 134 / 84 mmHg
Last Encounter Record: 10-Nov-2025 (10 months ago) (Grade: Grade 1 (Mild NPDR))

AI DR Severity: Moderate NPDR (Conf: 41.2%)



RECOMMENDED REFERRAL CARE: Kolar District Hospital - Ophthalmic Division [Level 2: Secondary District Eye Hospital]

Est. Distance: ~28.5 km | **Timeline:** SECONDARY REFERRAL WITHIN 30 DAYS | **Helpline:** +91 8152 222 344 (District Eye OPD)

OPHTHALMOLOGIST TELEMEDICINE REVIEW & VALIDATION (Target Review Time: under 30 Seconds)

Clinical Impression: [] Concordant with AI diagnosis [] Upgrade Severity [] Downgrade / Artefact
Prescribed Action: [] Tele-Consultation Booked [] Anti-VEGF / Laser Referral [] PHC Annual Rescreen
Reviewing Ophthalmologist Signature: Date/Time: